

1. What is the difference between WHERE and HAVING?

Answer: WHERE filters rows before grouping; HAVING filters groups after GROUP BY, so it can use aggregates.

2. Find the second highest salary from the employees table.

```
SELECT MAX(salary) AS second_highest  
FROM employees  
WHERE salary < (SELECT MAX(salary) FROM employees);
```

3. Find the second lowest salary from the employees table.

```
SELECT MIN(salary) AS second_lowest  
FROM employees  
WHERE salary > (SELECT MIN(salary) FROM employees);
```

4. What is the difference between ROW_NUMBER, RANK and DENSE_RANK?

Answer: ROW_NUMBER gives 1,2,3 even for ties; RANK gives ties the same number and leaves gaps (1,1,3); DENSE_RANK gives ties the same number without gaps (1,1,2).

5. What is a broadcast join in Spark?

Answer: The small table is copied to every executor so the large table does not need to be shuffled. Use it when one side comfortably fits in memory.

6. Why is data skew a problem in Spark?

Answer: A few keys hold most rows, so one task runs far longer than the rest. Fix with AQE skew join, salting, or broadcasting the smaller side.

7. Explain the Medallion architecture.

Answer: Bronze keeps raw data as received, Silver holds cleaned and de-duplicated data, and Gold holds business-ready aggregates for BI and ML.